

Upper Division Tutoring Program Topic Review

Facilitators: Jackie Jones

Raphael Bajet

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Math 110 Midterm Exam Review

Basic Definitions, Theorems, and Reference

Remark. In Math 110, a field will always refer to either $\mathbb{R}$ or $\mathbb{C}$.

Definition 1 (Linear Map). Let V, W be vector spaces over a field $\mathbb{F}$. A linear map $T : V \rightarrow W$ is a map that satisfies the following properties:

- For all $v, w \in V$, $T(v) + T(w) = T(v + w)$, and
- for all $\lambda \in \mathbb{F}$, $\lambda T(v) = T(\lambda v)$.

Definition 2 (Conditions for a subspace). A subset U of V is a subspace of V if and only if U satisfies the following three conditions.

- Additive Identity ($0 \in U$)
- Closed Under Addition ($u, w \in U \implies u + w \in U$)
- Closed Under Scalar Multiplication ($a \in \mathbb{F}, u \in U \implies au \in U$)

Definition 3 (Linear Independence). A list of vectors $v_1, v_2, \dots, v_m$ in V is called *linearly independent* if the only choice of scalars $a_1, a_2, \dots, a_m \in \mathbb{F}$ that makes

$$a_1 v_1 + a_2 v_2 + \dots + a_m v_m = 0$$

is $a_1 = a_2 = \dots = a_m = 0$. The empty list $()$ is also declared to be linearly independent.

Definition 4 (Basis and Dimension). Let V be a vector space. A basis of V is a linearly independent list of vectors $\{v_1, \dots, v_n\}$ that spans the space, i.e. any vector in V can be written as a unique linear combination of the v_i 's. The dimension of the space is defined to be the length of any basis, e.g. $\dim(V) = n$.

Definition 5 (Change-of-Basis Formula). Suppose $T \in \mathcal{L}(V)$, and that $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are bases of V . Let $A = \mathcal{M}(T, (u_1, \dots, u_n))$ and $B = \mathcal{M}(T, (v_1, \dots, v_n))$ and $C = \mathcal{M}(I, (u_1, \dots, u_n), (v_1, \dots, v_n))$. Then $A = C^{-1}BC$.

Definition 6 (Sum, Direct Sum). For subspaces U, W of V , their sum $U + W = \{u + w : u \in U, w \in W\}$ is also a subspace of V . Note $\dim U + \dim W \geq \dim U + W$ except when $U \cap W = 0$ in which case $U \oplus W = U + W$ is a direct sum.

Definition 7 (Null Space and Range). Let $T : V \rightarrow W$ be a linear map. The null space (also known as the kernel) of T is the set of vectors $\text{Null}(T) = \{v \in V : T(v) = 0\}$ and is a subspace of V . The range of T is the set of vectors $\text{Range}(T) = \{w \in W : \text{there exists } v \in V \text{ such that } T(v) = w\}$ and is a subspace of W .

Theorem 1 (Fundamental Theorem of Linear Maps). Suppose V is finite dimensional and $T : V \rightarrow W$ is a linear map. Then $\dim(V) = \dim(\text{Null}(T)) + \dim(\text{Range}(T))$.

Definition 8 (Isomorphism Between Vector Spaces). An isomorphism between two vector spaces is an invertible linear map. Moreover, two finite-dimensional vector spaces over $\mathbb{F}$ are isomorphic if and only if they have the same dimension.

Theorem 2. A linear map is invertible if and only if it is injective and surjective.

Theorem 3. Suppose V and W are finite-dimensional. Then $\mathcal{L}(V, W)$ is finite-dimensional and $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$.

Definition 9 (Linear Functional). A linear functional on V is a linear map from V to $\mathbb{F}$.

Definition 10 (Dual Space). The dual space of V , otherwise known as $V' = \mathcal{L}(\mathbb{F}, V)$, is the vector space of all linear functionals on V .

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Definition 11 (Dual Basis). If $v_1, \dots, v_n$ is a basis of V , then the dual basis of $v_1, \dots, v_n$ is a list of linear functionals $\varphi_j \in V'$ such that

$$\varphi_j(v_k) = \begin{cases} 1 & \text{if } k = j, \\ 0 & \text{if } k \neq j. \end{cases}$$

Definition 12 (Dual Map). Suppose $T : V \rightarrow W$ is a linear map between finite dimensional spaces over a field $\mathbb{F}$. We can define a map $T' : W' \rightarrow V'$ between their respective dual spaces via $T'(\varphi) = \varphi \circ T$. Note that $\varphi \circ T$ is indeed a linear map from V to $\mathbb{F}$ and therefore belongs to V' .

Remark. If V is finite-dimensional, V' has the same dimension so they are isomorphic, but not ‘canonically’ so, i.e. different isomorphisms arise from picking different bases. However, $\Phi : V \rightarrow V''$ given by $\Phi(v)(\varphi) = \varphi(v)$ is canonical; note we never have to choose a basis.

Definition 13 (Annihilator). For a subspace U of V , the annihilator of U , $U^0 = \{\varphi \in V' : \varphi(u) = 0 \forall u \in U\}$, is the set of functionals that send every $u \in U$ to 0.

Theorem 4. The following properties are instrumental to many proofs relating V and V' :

$$\text{null } T' = (\text{range } T)^0 \quad \text{range } T' = (\text{null } T)^0$$

Section II: Exercises

1. Suppose that U is a subspace of a finite-dimensional vector space V . Prove that there is a subspace W of V such that $U \oplus W = V$.
2. Let $T : P_3(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ be the map $T(p) = \frac{d^2}{dx^2}p + \frac{d}{dx}p$. For example $T(x^3 + 2) = 6x + 3x^2$. Compute the matrix $\mathcal{M}(T)$ with respect to the standard basis of $P_3(\mathbb{R})$ and $P_2(\mathbb{R})$.
3. Let V be the subspace of $\mathbb{P}_2$ of polynomials of degree less than or equal to 3 such that $p'(1) = p(2)$. Construct a basis of V and prove that it is indeed a basis.
4. Let V and W be two finite-dimensional vector spaces over a field $\mathbb{F}$. Prove that if V and W are isomorphic, then they have the same dimension.
5. Does there exist a linear map $T : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ such that

$$\text{null } T = \{(x, x, x, x) \in \mathbb{R}^4 | x \in \mathbb{R}\}?$$

6. Prove that the intersection of subspaces is a subspace. What about union?
7. Suppose that V is a finite-dimensional vector space with $T \in \mathcal{L}(V)$, and fix a basis $v_1, \dots, v_n$ of V . Prove T is an isomorphism if and only if $Tv_1, \dots, Tv_n$ are linearly independent.
8. Give an example of a map $T \in \mathcal{L}(\mathbb{R}^2)$ with $\mathbb{R}^2 = \ker T \oplus \text{range } T$.
9. Say S, T are operators on V with $\text{range } S \subset \ker T$, show $(ST)^2 = 0$.
10. Say $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}), \mathcal{P}_2(\mathbb{R}))$ is the differentiation map $D(p(x)) = p'(x)$. Find bases α, β such that

$$\mathcal{M}_{\alpha, \beta}(D) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

11. Let V be a real vector space, and let $P : V \rightarrow V$ be a linear map satisfying $P^2 = P$. Such a linear map is called a projection. Show that $V = \text{null } (P) \oplus \text{range } (P)$.
12. (Sp26 Practice Midterm, Q7) Let V and W be two finite-dimensional vector spaces over a field $\mathbb{F}$, let $T : V \rightarrow W$ be a linear map, and $T' : W' \rightarrow V'$ the dual map. Suppose that T is surjective. Prove that T' is injective.
13. (Axler 3F.22) Suppose V is finite-dimensional and U and W are subspaces of V . Prove (a) $(U + W)^0 = U^0 \cap W^0$ and (b) $(U \cap W)^0 = U^0 + W^0$